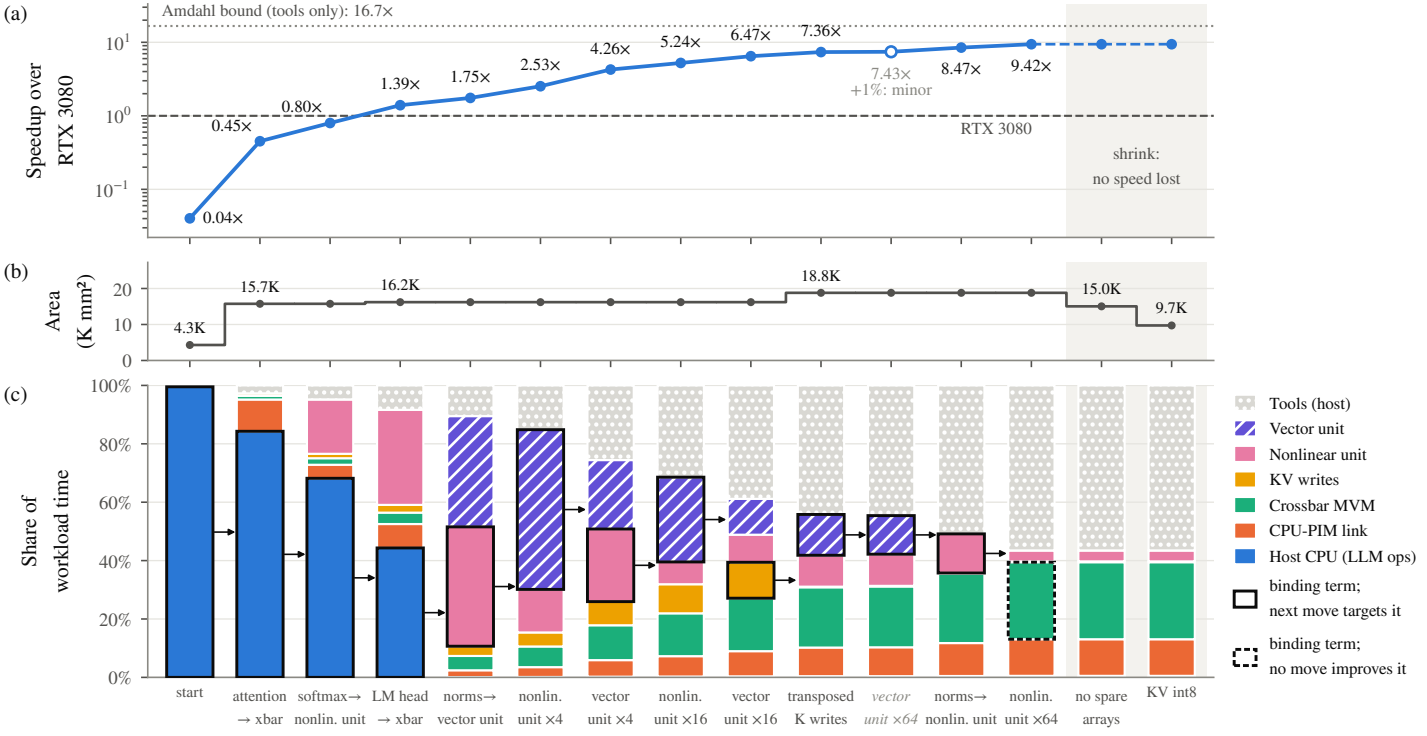

(a)
Amdahl bound (tools only): 16.7×
Speedup over RTX 3080
10^1
10^0
10^{-1}
0.04×
0.45×
0.80×
1.39×
1.75×
2.53×
4.26×
5.24×
6.47×
7.36×
7.43×
+1%: minor
8.47×
9.42×
RTX 3080
shrink:
no speed lost
(b)
Area
(K mm²)
20
10
0
4.3K
15.7K
16.2K
18.8K
15.0K
9.7K
(c)
Share of
workload time
100%
80%
60%
40%
20%
0%
start
attention → xbar
softmax→ nonlin. unit
LM head → xbar
norms→ vector unit
nonlin. unit ×4
vector unit ×4
nonlin. unit ×16
vector unit ×16
transposed K writes
vector unit ×64
norms→ nonlin. unit
nonlin. unit ×64
no spare arrays
KV int8
Tools (host)
Vector unit
Nonlinear unit
KV writes
Crossbar MVM
CPU-PIM link
Host CPU (LLM ops)
binding term; next move targets it
binding term; no move improves it